

Knowledge Distillation Across Teacher Architectures for Edge-Efficient Plant Disease Classification

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Abstract. Accurate image-based plant disease classification can support precision agriculture, but high-capacity networks are not always convenient for edge or mobile deployment. This work studies knowledge distillation from large teacher networks into a compact MobileNetV3-Small student, and asks in particular whether the *architecture of the teacher* affects the distilled student. As a preliminary demonstration, we distill a ResNet50 teacher into MobileNetV3-Small on the 38-class PlantVillage dataset with a single train/validation/test split; the distilled student reaches 0.9937 accuracy and 0.9907 macro-F1 with 1.56M parameters, about 15 times fewer than the teacher. We then carry out the main study on the smaller and class-imbalanced Cassava leaf-disease dataset (five classes) using stratified five-fold cross-validation. Three teachers from different architectural families — ResNet50 (residual CNN), EfficientNet-B2 (compound-scaled CNN), and ViT-B/16 (vision transformer) — are each retrained per fold and distilled into the same student, and we report accuracy together with macro-averaged precision, recall, and F1. On Cassava, distillation improves the student over a supervised baseline by 1.4–1.8 percentage points of accuracy, but the choice of teacher has little effect: the three distilled students agree to within 0.5 points despite the teachers differing widely in accuracy (82.4–85.1%) and in size (7.7M–85.8M parameters). These results suggest that the benefit of distillation comes from the procedure itself rather than from the capacity or architecture of the teacher.

Keywords: Knowledge distillation · Plant disease classification · MobileNetV3 · Cross-validation · Vision transformer · Edge inference.

1 Introduction

Plant disease recognition is a practical computer vision problem with direct relevance to crop monitoring and precision agriculture. Deep neural networks can achieve strong classification accuracy on curated leaf image datasets, but models with high parameter counts may be inconvenient for mobile or edge-oriented deployment. This creates a tension between predictive accuracy and computational efficiency.

This work studies that trade-off through knowledge distillation, in which a high-capacity teacher network guides the training of a lightweight MobileNetV3-Small student. The aim is not to claim deployment on a specific device, but to evaluate whether a compact student can retain a teacher’s predictive behavior while using far fewer parameters. Beyond the standard teacher-to-student comparison, we focus on a question that a single experiment cannot answer: does the *choice of teacher architecture* change how well the student learns?

We make three contributions: (i) a preliminary single-split distillation study on 38-class PlantVillage confirming that distillation improves a compact student; (ii) a main cross-validated study on Cassava comparing three teacher architectures (ResNet50, EfficientNet-B2, ViT-B/16) under stratified five-fold cross-validation, with the teacher retrained per fold and results reported as mean $\pm$ standard deviation of accuracy, macro-averaged precision, recall, and F1 with paired significance tests; and (iii) a post-training quantization analysis of the distilled student as an additional edge-compression step.

2 Related Work

PlantVillage [8] provides a widely used benchmark of labeled plant leaf images. Its clean imaging conditions are well suited to initial controlled experiments but less representative of field conditions. The Cassava leaf-disease dataset [11] covers five classes with strong class imbalance and real field imagery, providing a harder and more realistic testbed. Early work demonstrated high accuracy on PlantVillage with convolutional networks [10], but the literature has paid comparatively little attention to compressing these models for on-device inference.

Knowledge distillation [6] transfers information from a larger teacher to a smaller student by having the student match the teacher’s softened output probabilities. Numerous extensions exist; a broad survey is given by Gou et al. [4]. We use the original logit-based formulation since our focus is the teacher architecture rather than the distillation mechanism.

A recurring finding is that stronger teachers do not necessarily produce better students. Cho and Hariharan [2] show that a large capacity gap can impair distillation, and Mirzadeh et al. [9] propose intermediate teacher assistants to bridge it. Beyer et al. [1] argue that training consistency matters more than teacher accuracy. These results motivate our central question: when the student is fixed, does the teacher’s *architecture* — and not merely its accuracy — affect the distilled student? DeiT [13] distills *into* a transformer using a CNN teacher; our setting is complementary, distilling *from* a transformer and two CNN teachers into a convolutional student.

The three teachers span distinct architectural families: ResNet50 uses residual connections to ease optimization of deeper convolutional models [5]; EfficientNet-B2 applies compound scaling of depth, width, and resolution [12]; and ViT-B/16 processes images as patch sequences via self-attention [3]. The student in all experiments is MobileNetV3-Small [7], a compact architecture designed for mobile inference.

3 Methodology

Let z_t and z_s denote the teacher and student logits for an input image, and let y denote the ground-truth class label. Supervised teacher and supervised student baselines are trained with cross-entropy against the hard labels. For distillation, the teacher is first trained and then frozen, and the student is optimized using a weighted sum of the cross-entropy loss and a temperature-scaled Kullback–Leibler divergence term:

$$\mathcal{L} = \alpha T^2 \text{KL}\left(\text{softmax}\left(\frac{z_t}{T}\right) \parallel \text{softmax}\left(\frac{z_s}{T}\right)\right) + (1 - \alpha) \text{CE}(y, z_s). \quad (1)$$

The T^2 factor compensates for the gradient scaling introduced by softened probabilities. The temperature is fixed at $T = 4.0$ throughout; the distillation weight is $\alpha = 0.5$ in the PlantVillage study and $\alpha = 0.7$ in the main Cassava study. In both cases α is held constant across teachers so that any difference between teachers is attributable to the teacher rather than to the loss weighting.

4 Experimental Setup

All experiments use pretrained ImageNet weights, images resized to 224×224 pixels and normalized with ImageNet statistics, random horizontal flipping and small rotations during training, deterministic resizing for validation and test, and the best validation checkpoint for evaluation. Latency is measured as a batch-one average over repeated forward passes on local hardware and should not be read as a deployment benchmark.

4.1 Main Study: Cassava Cross-Validation

The main study uses the Cassava leaf-disease dataset (five classes) under stratified five-fold cross-validation. For each fold, the teacher is retrained from scratch on that fold’s training partition and then frozen before distilling the student; this avoids the bias that would arise if a single teacher had seen images that later appear in the test fold. The fold partition is fixed by seed, so every model sees identical folds. Three teachers are evaluated — ResNet50, EfficientNet-B2, and ViT-B/16 — each distilled into MobileNetV3-Small and compared against a supervised baseline; per-fold training schedules are given in Table 1. Results are reported as mean $\pm$ standard deviation across the five folds of accuracy and macro-averaged precision, recall, and F1. Each distilled student is compared against the baseline with a paired two-sided t -test on per-fold accuracies. All runs use an NVIDIA RTX 4070 with automatic mixed precision.

4.2 Preliminary Study: PlantVillage Single Split

As a preliminary demonstration, a ResNet50 teacher, a supervised MobileNetV3-Small baseline, and a distilled student are trained on the full 38-class PlantVillage

Table 1. Per-fold training configuration for the Cassava cross-validation study. All teachers are distilled into MobileNetV3-Small with $T = 4.0$ and $\alpha = 0.7$.

Model	Role	Learning rate	Epochs	Batch
ResNet50	Teacher	3×10^{-4}	10	64
EfficientNet-B2	Teacher	3×10^{-4}	10	64
ViT-B/16	Teacher	1×10^{-4}	10	32
MobileNetV3-Small	Student / baseline	5×10^{-4}	15	64

dataset with a single 70/15/15 train/validation/test split (seed 42), 15 epochs, batch size 32, $T = 4.0$, $\alpha = 0.5$, on Apple Silicon (MPS). Learning rates are 3×10^{-4} (teacher), 10^{-3} (supervised student), and 5×10^{-4} (distilled student).

5 Results

Experiments cover two datasets. On PlantVillage (38 classes, single split), we train one ResNet50 teacher, one supervised MobileNetV3-Small baseline, and one distilled student. On Cassava (five classes, five-fold CV), we train three teacher baselines, one student baseline, and three distilled students — one per teacher — plus an ablation over distillation hyperparameters and two post-training quantization variants. All metrics are macro-averaged precision, recall, and F1 alongside accuracy.

5.1 Cassava Cross-Validation

Table 2 summarizes the cross-validated results. Two questions are of interest: whether distillation improves the student over the supervised baseline on this harder, imbalanced dataset, and whether a stronger or architecturally different teacher distills better than ResNet50.

Table 2. Cross-validated test results on cassava (5-fold, mean $\pm$ standard deviation across folds). Precision, recall, and F1 are macro-averaged. The teacher is retrained per fold; the student and KD hyperparameters are fixed.

Teacher	Student	Accuracy (%)	Precision (%)	Recall (%)	Macro-F1 (%)	Parameters
–	resnet50	85.13 ± 0.95	75.08 ± 1.40	73.69 ± 1.28	74.00 ± 0.78	23,518,277
–	efficientnet_b2	84.59 ± 0.67	73.91 ± 1.51	71.75 ± 0.86	72.60 ± 0.61	7,708,039
resnet50	mobilenet_v3_small	83.49 ± 0.31	72.70 ± 0.66	69.74 ± 2.62	70.72 ± 1.99	1,522,981
vit_b_16	mobilenet_v3_small	83.12 ± 0.72	72.90 ± 0.99	68.25 ± 2.36	69.85 ± 1.95	1,522,981
efficientnet_b2	mobilenet_v3_small	83.04 ± 0.41	71.75 ± 1.53	69.40 ± 1.98	70.34 ± 1.26	1,522,981
–	vit_b_16	82.40 ± 0.64	71.41 ± 1.28	69.08 ± 1.58	69.38 ± 1.26	85,802,501
–	mobilenet_v3_small	81.67 ± 1.01	69.83 ± 2.64	67.57 ± 1.38	68.22 ± 0.80	1,522,981

Distillation helps. All three teachers raise the student above the supervised baseline (81.67% accuracy) by a similar margin: +1.82 points for ResNet50

(83.49%), +1.45 for ViT-B/16 (83.12%), and +1.37 for EfficientNet-B2 (83.04%). In the paired t -test the ResNet50 and ViT-B/16 gains are statistically significant ($p = 0.029$ and $p = 0.036$); the EfficientNet-B2 gain does not reach significance ($p = 0.079$), most likely reflecting the limited power of five folds and that run’s larger fold-to-fold variance.

The more notable result is how little the teacher matters. The three distilled students lie within 0.5 points of one another (83.04–83.49%), a spread no larger than their across-fold standard deviations, even though the teachers differ substantially in accuracy (82.40–85.13%), architectural family (two CNNs and one transformer), and size (7.7M, 23.5M, 85.8M parameters). The ViT-B/16 student (83.12%) surpasses its own teacher (82.40%), which runs counter to the expectation that a very large teacher, separated from a 1.5M- parameter student by a wide capacity gap, would distill poorly. These observations indicate that the improvement comes from the softened-label regularization of distillation rather than from the specific teacher — a practically convenient result, since a smaller and cheaper teacher suffices.

5.2 Sensitivity to Distillation Hyperparameters

The main study fixes $T = 4.0$ and $\alpha = 0.7$ across teachers. To verify this choice is not an artifact, we run a single-split sensitivity analysis for the ResNet50 teacher on fold 0, sweeping $T \in \{1, 2, 4, 8\}$ at $\alpha = 0.7$ and $\alpha \in \{0.3, 0.5, 0.7, 0.9\}$ at $T = 4.0$.

Table 3. Single-split KD sensitivity analysis for the resnet50 teacher (fold 0 of the Cassava cross-validation split). Temperature is swept at $\alpha = 0.7$ and α at $T = 4$; † marks the default used in the main study. Precision, recall, and F1 are macro-averaged. Values are single-split point estimates (no fold error bars).

Sweep	T	α	Accuracy (%)	Precision (%)	Recall (%)	Macro-F1 (%)
temperature 1	0.7		83.50	73.04	70.05	71.35
temperature 2	0.7		83.79	74.38	69.01	71.29
temperature 4	0.7 [†]		84.14	73.99	70.29	71.52
temperature 8	0.7		83.67	73.26	70.03	71.37
alpha	4	0.3	82.78	70.59	70.76	70.61
alpha	4	0.5	83.60	72.91	69.64	70.89
alpha	4	0.7 [†]	84.14	73.99	70.29	71.52
alpha	4	0.9	83.11	73.01	67.04	69.39

The student is stable: accuracy stays within one point over all settings (82.78–84.14%), with a gentle maximum at the default $T = 4.0$, $\alpha = 0.7$. Setting $\alpha = 0.9$ lowers macro-F1 to 69.39%, consistent with over-weighting the teacher at the expense of minority classes. The flat response confirms that the fixed configuration is well justified.

5.3 Edge Compression: Quantization

We apply two int8 post-training quantization (PTQ) strategies to the single-split distilled student on CPU using the `fbgemm` backend: *dynamic* PTQ (weights only) and *static* PTQ (weights and activations, calibrated on 256 training images). Table 4 reports the accuracy/size/latency trade-off.

Table 4. Post-training quantization of the distilled MobileNetV3-Small on Cassava (single-split test set, CPU, `fbgemm`). Precision, recall, and F1 are macro-averaged.

Variant	Accuracy (%)	Precision (%)	Recall (%)	Macro-F1 (%)	Size (MB)	Latency (ms)
Float32 (baseline)	84.23	74.31	70.42	71.87	5.93	17.7
Dynamic int8	84.20	74.32	70.35	71.83	4.23	16.0
Static int8	12.22	21.06	20.43	6.05	1.74	18.4

Dynamic quantization is essentially lossless ($84.23 \rightarrow 84.20\%$, a 0.03-point change) and provides a free $1.4\times$ size reduction. Static quantization reaches $3.4\times$ compression but collapses accuracy to 12.2%, well below the 20% random baseline. This collapse reflects the known sensitivity of MobileNetV3-Small’s depthwise-separable convolutions and Hardswish activations to post-training quantization; quantization-aware training would be required to recover accuracy at full int8 compression, which we leave to future work.

5.4 PlantVillage Single Split

Table 5 reports the single-split results on the 38-class PlantVillage task. The improvement from 0.9851 to 0.9937 accuracy after distillation shows that the teacher logits provide useful signal beyond the hard labels, even when the supervised student is already strong. This limited headroom motivates the main cross-validated study on the harder Cassava data.

Table 5. Single-split test results on the 38-class PlantVillage task. Precision, recall, and F1 are macro-averaged.

Model	Training strategy	Accuracy	Precision	Recall	Macro-F1	Parameters	Latency
ResNet50	Teacher supervised	0.9936	0.9920	0.9908	0.9912	23,585,894	5.47 ms
MobileNetV3-Small	Supervised baseline	0.9851	0.9788	0.9781	0.9779	1,556,806	3.51 ms
MobileNetV3-Small	Distilled from ResNet50	0.9937	0.9911	0.9907	0.9907	1,556,806	3.53 ms

5.5 Per-Class Analysis on Cassava

The gap between accuracy and macro-F1 throughout the Cassava results is a direct consequence of class imbalance. On the single-split test set, *mosaic disease*

accounts for 1,984 of 3,209 images (62%), whereas *bacterial blight* has only 155 (5%). Table 6 and Figure 1 give the per-class performance and confusion matrix of the distilled student.

Table 6. Per-class performance of the distilled MobileNetV3-Small on the Cassava single-split test set.

Class	Precision Recall		F1 Support	
Bacterial blight	0.667	0.465	0.548	155
Brown streak disease	0.793	0.697	0.742	307
Green mottle	0.707	0.701	0.704	378
Healthy	0.616	0.712	0.660	385
Mosaic disease	0.934	0.947	0.940	1,984

The student performs well on the dominant class (mosaic disease, F1 0.94) but degrades on minority classes, reaching its worst on bacterial blight (F1 0.55, recall 0.47). Because accuracy is dominated by the majority class while macro-F1 weights all classes equally, accuracy (84%) substantially overstates per-class competence (macro-F1 72%), which is why we report both throughout. The confusion matrix reveals two systematic error patterns: minority diseases are frequently predicted as *healthy* (59 of 155 bacterial-blight leaves, 38%), and the visually similar *green mottle* and *mosaic disease* classes are mutually confused. Addressing these errors would likely require class re-balancing or cost-sensitive training.

6 Conclusion

This work studied knowledge distillation from large teachers into a compact MobileNetV3-Small student for edge-efficient plant disease classification. On PlantVillage, a ResNet50 teacher distills into a student with 0.9937 accuracy at about 15× fewer parameters. The main cross-validated study on Cassava compared three teacher architectures — ResNet50, EfficientNet-B2, and ViT-B/16 — and found that distillation consistently raises the student by 1.4–1.8 accuracy points over the supervised baseline. Crucially, the choice of teacher had little effect: all three distilled students agreed to within 0.5 points despite teachers differing widely in accuracy, architecture, and size, and the weakest and largest teacher (ViT-B/16) taught as effectively as the strongest. The benefit of distillation in this setting therefore appears to stem from the softened-label regularization rather than from the specific teacher, which is encouraging for edge deployment because a small, inexpensive teacher suffices. Dynamic int8 post-training quantization provides a further, orthogonal 1.4× size reduction at no accuracy cost, though full int8 compression requires quantization-aware training to avoid accuracy collapse.

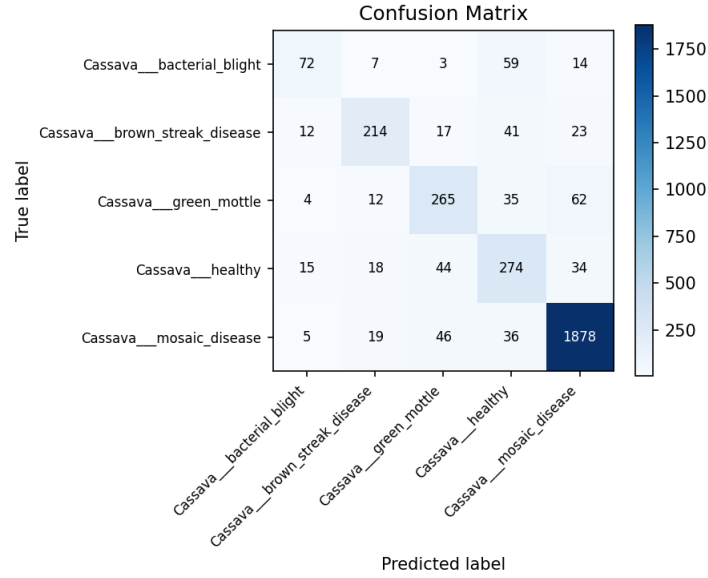


Fig. 1. Confusion matrix of the distilled MobileNetV3-Small on the Cassava single-split test set. Errors concentrate in the minority classes.

Several limitations remain. Distillation hyperparameters are held fixed across teachers, and with only five folds statistical power is limited. Both datasets are curated, so performance on field images with cluttered backgrounds and varying lighting may differ, and latency is measured on local hardware rather than target edge devices.

Disclosure of Interests. The authors have no competing interests to declare.

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